

Real-time dynamic simulation of angular velocity and suppression of dead zone in IFOG

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Abstract The mechanism of the dead zone in closed-loop IFOGs is theoretically analyzed. The digital closed-loop transfer model for fiber-optic gyroscope is modified and the electronic cross-coupling interference from the feedback channel in the output signal of the detector in the forward channel is added. Using this model, we achieve a real-time dynamic simulation for the dead zone. The simulation results are basically consistent with experiment results. Before the electronic cross-coupling has been eliminated, the dead zone of the high-precision fiber-optic gyro is about $0.04^{\circ}/h$, while the medium precision fiber-optic gyro is $0.4^{\circ}/h$. After modifying the modulator circuit boards, the dead zone and the noise of the IFOG are of the same order of magnitude, better than $0.001^{\circ}/h$ and $0.01^{\circ}/h$, respectively.

Keywords Dead zone · Closed-loop transfer model · Real-time dynamic simulation · Electronic cross-coupling

1 Introduction

Optical gyros have a number of advantages over their mechanical counterpart based on either conservation of angular momentum or the Coriolis force. They operate on the principle of the Sagnac effect [1]. The fiber-optic gyros (FOGs) belong to optical gyros whose sensitive elements are the fiber-optic coils [2]. Because of their advantages such as totally solid state, great reliability, low cost and rapid reaction [3–7], the FOGs become one of the most

important optical gyros, which are extensively applied to airplanes, submarines, missiles, etc. [8, 9]. Based on their principles, the FOGs can be roughly divided into the interferometric fiber-optic gyroscopes (IFOGs), the resonant fiber-optic gyroscopes (RFOGs) and others. Currently, the IFOGs are widely used, while the RFOGs are at the stage of laboratory research.

Along with the improvement of the precision, the advantages of IFOGs gradually emerge and they will replace more expensive electrostatic gyroscopes in the near future. The dead zone is a special area, in which the gyro is insensitive to the input angular velocity, and its magnitude determines the precision of the whole system. Theoretically, the dead zone and the noise of the IFOGs are of the same orders of magnitude. But, actually in specific cases, the dead zone is about several orders of magnitude larger than the noise as a result of all kinds of crosstalk and modulation errors in the circuit boards.

The study of dead zone was once reported before [10–12]. We consider that the dead zone is mainly caused by the electronic cross-coupling interference between the signals of feedback channel and forward channel. On this basis, we establish the digital closed-loop transfer model including the electronic cross-coupling interference and realize a real-time dynamic simulation of the dead zone. Different from the method mentioned in the report, which introduced a random voltage signal on the Y waveguide phase modulator [11], we restrain the dead zone by diminishing the electronic cross-coupling interference according to the results of our simulation. The approach we diminish the interference mainly includes wiring the analog circuit and the digital circuit separately, strengthening the power supply filter, optimizing the layout of ground wire and taking the digital step wave along with the analog square wave to modulate the signal. We basically eliminate

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the dead zone with our improvement. We measure the dead zone of the fiber-optic gyroscope before and after restraining the cross-coupling interference, respectively. The outcomes reflect that the dead zone becomes larger when the electronic cross-coupling interference exists. As it should be, the dead zone and the noise of the IFOGs are of the same order of magnitude when the electronic cross-coupling interference was effectively restrained. The outcomes of our simulation basically prove our theoretical analysis.

2 The generation mechanism of dead zone in the digital closed-loop IFOGs

High-precision IFOGs always operate in the digital closed-loop mode, which has the advantages of wider dynamic range and greater linearity of the scaling factor [13–16].

Figure 1 shows the schematic diagram of the digital closed-loop IFOGs. The light power detected by the detector can be written as

$$P = P_0(1 + \cos \phi_S), \quad (1)$$

where ϕ_S is the Sagnac phase shift, P_0 is the average optical power reaching the detector. As we know, when the input value is close to zero, the cosine function exhibits the lowest sensitivity. To increase the sensitivity, a square wave is applied at the phase modulator. The square wave is expressed as $\phi_m = \pm\phi_b$, where ϕ_b equals to half of the peak of the square wave, whose half cycle equals to the propagation time of the light in the fiber coil (τ). In the case of small rotation signal, the light power difference between the two adjacent modulation states in the case of open loop (without feedback) can be written as:

$$\Delta P = 2P_0 \sin \phi_b \sin \phi_S \approx 2P_0 \phi_S, \quad (2)$$

in which, we have assumed $\phi_b = \pi/2$. This optical signal is converted into a digital quantity after photoelectric

conversion, pre-amplification and A/D conversion. The digital quantity is stored in a register and used as the output rotation signal of the gyro. This digital quantity becomes a digital step signal after accumulating, whose duration of each digital step equals to τ . The superposition of the digital step wave and the square wave is converted into an analog voltage signal by a D/A converter. This analog voltage signal is applied to the integrated optical phase modulator, and an additional phase shift in the fiber coil is introduced artificially. The relationship between the modulate voltage and the additional phase shift in the integrated phase modulator is:

$$\phi_M = K_M \times V_M = \phi_{\text{square}} + \phi_{\text{step}}, \quad (3)$$

where K_M is the modulation coefficient, V_M is the superposition analog voltage of the step wave and the square wave, ϕ_{square} is the phase shift introduced by the square wave, and ϕ_{step} is the phase shift introduced by the step wave. The digital step wave cannot be accumulated without limitation, after a certain period, it will generate a reset. In a digital closed-loop IFOG, the reset is generated automatically by the overflow of the D/A converter. If we adjust the gain of the modulation channel, when meet

$$2^{n_{\text{dac}}} \times V_{\text{LSB}} = 2V_\pi, \quad (4)$$

the overflow of the D/A converter automatically generates a 2π phase reset in the phase modulator. In Eq. (4), n_{dac} is the number of bits of the D/A converter, V_{LSB} is the analog voltage corresponding to the lowest effective bit of the D/A converter, and V_π is the analog voltage required to produce π phase shift in the integrated phase modulator. The waveforms of the phase shift in the integrated phase modulator generated by the D/A output analog voltage are shown in Fig. 2.

The additional phase difference between the two counter propagation light beams can be written as:

$$\phi_F = \phi_M(t) - \phi_M(t - \tau) = -\phi_S. \quad (5)$$

The amplitude of the compensation phase difference is equal to the Sagnac phase shift, while their sign is opposite. Thus, the IFOGs always work in the setting bias point. As shown in Fig. 2, when the gyro is static, the demodulated signal is equal to zero, and the step height of the phase step wave is also equal to zero. When the gyro rotates, the demodulated signal is no longer equal to zero, and the step height of the digital phase step wave equals to Sagnac phase shift.

The electronic cross-coupling, coming from the feedback channel, exists in the output signal of the detector in the forward channel (see Fig. 5, Sect. 3). It can be written as $\Delta V_c = K_c \times V_M$, where K_c is the electronic cross-coupling coefficient. This error signal can produce a phase error in the fiber coil after pre-amplification, A/D

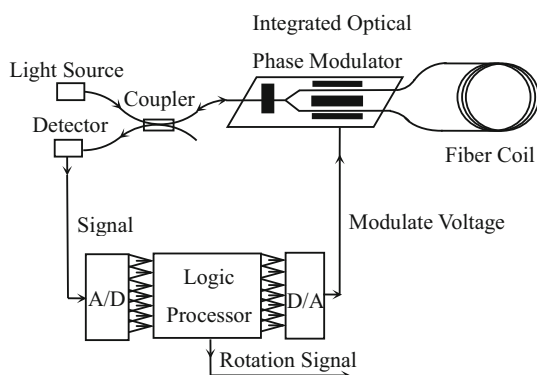


Fig. 1 Schematic diagram of the digital closed-loop IFOGs

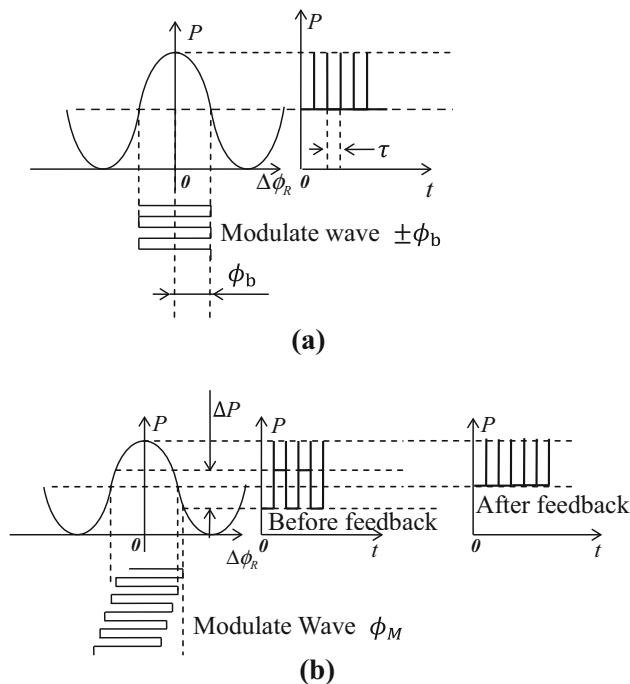


Fig. 2 Modulate wave and the optical signal reach the detector **a** the FOG is static; **b** the FOG is rotating

conversion, digital signal processing, and D/A conversion. The phase error difference between the two counter propagation light beams, introduced by the electronic cross-coupling can be written as:

$$\phi_e = K_M G' K_c \times [V_M(t) - V_M(t - \tau)], \quad (6)$$

where G' is the gain when the signal passes through the preamplifier, A/D converter, digital processor, and D/A converter. In general condition $G' = 1$, Eq. (4) can be written as:

$$\phi_e = K_c \times [\phi_M(t) - \phi_M(t - \tau)]. \quad (7)$$

If $\phi_M(t) = \phi_b + \phi_{\text{step}}$, then $\phi_M(t - \tau) = -\phi_b + \phi_{\text{step}} - \phi_F$, $\phi_e = K_c(2\phi_b + \phi_F) \approx 2K_c\phi_b$, considering the 2π phase reset in the phase modulator, ϕ_e can be written as:

$$\phi_e = \begin{cases} \phi_A = 2K_c\phi_b, & \phi_{\text{step}} < 2\pi - \phi_b, \\ \phi_B = K_c(2\phi_b - 2\pi), & \phi_{\text{step}} \geq 2\pi - \phi_b \end{cases} \quad (8)$$

where ϕ_A represents offset error during the step wave rising stage caused by the electronic cross-coupling, ϕ_B represents the offset error during the step wave reset stage.

As shown in Fig. 3a, during the rising and reset stage of the step wave, the step height and the slope are the same in the case of no electronic cross-coupling. Thus the gyro does not produce any error.

Figure 3b shows the schematic of the step wave in the case of electronic cross-coupling. During the rising and reset stage, the step height is $\phi_S + \phi_A$ and $\phi_S + \phi_B$ respectively. Compared with the ideal case in Fig. 3a, the

reset cycle of the step wave is changed and the output error of the gyro occurs. As long as $\phi_S + \phi_A > 0$, $\phi_S + \phi_B > 0$, the gyro still will not produce dead zone. For a given small Sagnac phase shift ϕ_S , in one reset cycle of the D/A converter, the duration of ϕ_A is $T_A = \frac{2\pi - \phi_b}{\phi_S + \phi_A} \times \tau$, the duration of ϕ_B is $T_B = \frac{\phi_b}{\phi_S + \phi_B} \times \tau$. The output signal of the gyro is the average value of the feedback signal within a certain period of time, which can be written as:

$$\begin{aligned} \bar{\phi} &= \frac{(\phi_S + \phi_A) \times \left(\frac{2\pi - \phi_b}{\phi_S + \phi_A} \tau \right) + (\phi_S + \phi_B) \times \left(\frac{\phi_b}{\phi_S + \phi_B} \tau \right)}{\frac{2\pi - \phi_b}{\phi_S + \phi_A} \tau + \frac{\phi_b}{\phi_S + \phi_B} \tau} \\ &= \frac{2\pi(\phi_S + \phi_A) \times (\phi_S + \phi_B)}{2\pi\phi_S + \phi_b\phi_A + (2\pi - \phi_b)\phi_B}. \end{aligned} \quad (9)$$

When $\phi_b = \pi/2$, we can get $\phi_B = -3\phi_A$, the above equation can be written as:

$$\bar{\phi} = \frac{(\phi_S + \phi_A) \times (\phi_S - 3\phi_A)}{\phi_S - 2\phi_A}. \quad (10)$$

In general condition, Eq. (10) does not equal to zero. But once $\phi_S + \phi_A > 0$ and $\phi_S + \phi_B < 0$ as shown in Fig. 3c, the D/A converter reset happens with great frequency. The duration of ϕ_A is $T_A = 2\tau$ the duration of ϕ_B is $T_B = -\frac{2(\phi_S + \phi_A)}{\phi_S + \phi_B} \tau$, and we can obtain $\bar{\phi} = 0$, so the gyro output is zero as shown in Fig. 4.

3 The dead zone simulation of the digital closed-loop IFOGs

The transfer model of the digital closed-loop fiber-optic gyroscope was initially used in the noise analysis of IFOGs. Then, it was used to calculate the performance parameters of IFOGs [17]. According to the different methods of signal processing, the closed-loop transfer model will be slightly different. According to the signal processing method we used, we established the closed-loop transfer model as shown in Fig. 5. In the schematic diagram, the ideal model is marked by the solid line and the dashed line indicates electronic cross-coupling interference of the feedback channel in the forward channel.

In the ideal conditions, the Z-transform of the digital output signal of IFOGs can be expressed as:

$$\begin{aligned} D_0(z) &= 2\eta K_I R_f G_a G_f K_{AD} \times (1/(1 - Z^{-1})) \\ &\quad \times (K_0 \Omega_{IN} - K_M(1 - Z^{-1})V_M), \end{aligned} \quad (11)$$

where $K_I = P_0 \sin \phi_b$, P_0 represents the average optical power reaching the detector, $K_0 = 2\pi LD/\lambda c$, in which L is the fiber length, D is the diameter of fiber coil, λ is the optical wavelength, and c is the light speed; η is the conversion efficiency of the detector; R_f is trans-impedance of the detector; G_a is the gain of the preamplifier; G_f is the

Fig. 3 Schematic of the phase step wave **a** no electronic cross-coupling; **b** including electronic cross-coupling, while $\phi_S + \phi_B > 0$; **c** including electronic cross-coupling, while $\phi_S + \phi_B < 0$

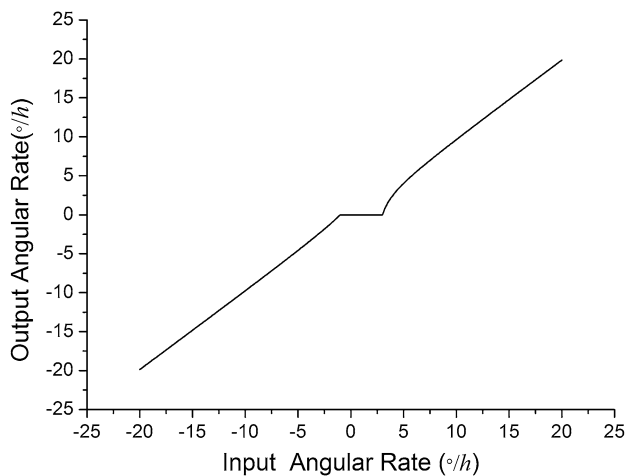
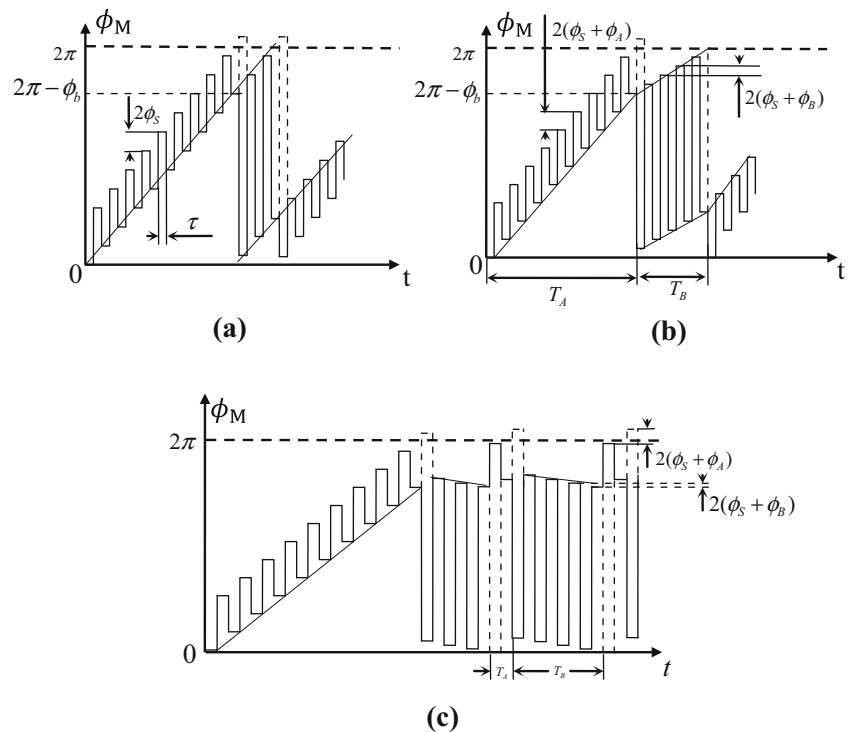


Fig. 4 Theoretical analysis of the dead band of IFOGs

filter gain of the preamplifier and $G_f \approx 1$; and $K_{AD} = 2^{n_{adc}}/V_{REF1}$ is the gain coefficient of the A/D converter, in which n_{adc} is the number of bits A/D converter and V_{REF1} is the reference voltage of the A/D converter. K_M is the modulation coefficient of phase modulator. V_M is the step wave voltage that adds to the phase modulator, whose Z-transform can be expressed as:

$$V_M(z) = G_P G_D K_{DA} \times \frac{1}{1 - Z^{-1}} \times D_0(z) - H(z) V_{2\pi}. \quad (12)$$

In the equation above, $K_{DA} = V_{REF2}/2^{n_{dac}}$ is the gain coefficient of the D/A converter. Here, V_{REF2} is the

reference voltage of the D/A converter, n_{dac} is the number of bits of the D/A converter. $G_D = \frac{1}{2^{n_1 - n_{dac}}}$, $G_P = 2V_\pi/V_{REF2}$ is the gain in the process of power amplification within the modulation channel. $H(z)$ means threshold function which can be expressed as:

$$H(z) = \begin{cases} 1 & V_M > 2V_\pi \\ 0 & V_M \leq 2V_\pi \end{cases}. \quad (13)$$

The relationship between the output rotation signal and the digital quantity of the gyroscope can be expressed as:

$$\Omega_0(z) = D_0(z) G_P K_{DA} K_M \times \frac{3600 \times 180}{2\pi \times K_0} (^\circ/h). \quad (14)$$

Considering the influence of electronic cross-coupling interference on the output signal from the detector, as is shown within the dashed part, combining Eq. (11), the Z-transform of the digital quantity of FOG can be expressed as:

$$D'_0(z) = \{2G_a G_f K_f \eta R_f [K_0 \Omega_{IN} - K_M (1 - Z^{-1}) V_M] - \Delta V_C\} \times K_{AD} \times (1/(1 - Z^{-1})), \quad (15)$$

where $\Delta V_C = K_C \times V_M$ is the electronic cross-coupling interference signal. Using Z inverse transform to Eq. (15), we can realize a real-time dynamic simulation for the input and output of gyroscope (Fig. 5). The typical parameters of the common-precision IFOG in our simulation are as follows: the length of optical fiber is 1000 m and the diameter of the fiber coil is 100 mm. The typical parameters of the high-precision IFOG are as follows: the length of the optical fiber is 3000 m and the diameter of the fiber coil

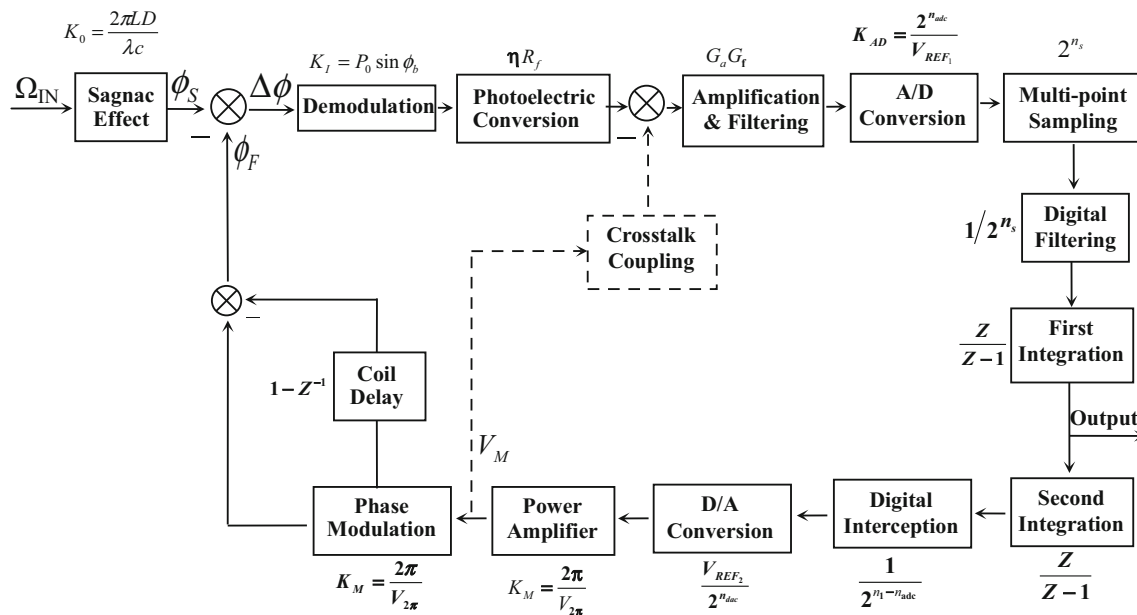


Fig. 5 Transfer model of the digital closed-loop IFOGs

200 mm. Other parameters are roughly the same. We set the $\lambda = 1550$ nm, $c = 3 \times 10^8$ m/s, $P_0 = 50$ μ W, $\eta = 0.8$, $R_f = 30$ K Ω , $n_{adc} = 11$, $V_{REF1} = 2$ V, $G_a = 10$, $G_f = 1$, $K_{DA} = 1/216$, $G_D = 1/24$, $V_{\pi} = 2$ V. The results of simulation are shown in Fig. 6.

As shown in Fig. 6, as a result of the electronic cross-coupling interference of the modulating channel in the demodulating channel, when the input angular rate is close to zero, the gyro output which is approximately zero is insensitive to the input. That is to say, there exists a dead zone. The region of the dead zone is straightly connected to the strength of electronic cross-coupling interference, the length of optical fiber and the diameter of the fiber coil. The longer and wider the optical fiber length and the diameter of the fiber coil are, the narrower the dead band is. When it is close to the dead zone, the gyro output will oscillate significantly. The closer it goes to the dead zone, the more obvious the oscillation is. The larger the dead zone is, the greater the oscillation amplitude becomes. In addition, the larger the dead zone, the greater error the gyro output between the true values is. All of these simulation results are basically consistent with our experimental results.

4 Experiment

We select two optical systems, one of them is used to assemble a medium precision gyroscope, the other is for a high-precision gyroscope. The elements of these two optical systems have the same performance except the fiber

length and the coil diameter. Each system consists of an ASE light source, a polarization-maintaining fiber coupler, a Y waveguide integrated optical phase modulator, a fiber coil, and a detector. We do the experiment using an original circuit board and a modified circuit board.

Figure 7 shows the experimental data. As the graphs show, increasing the fiber length and the coil diameter can reduce the dead zone. When it is close to the dead zone, the nonlinear phenomenon between the input and output of the gyro is serious. The dead zone of the high-precision fiber-optic gyro is $\sim 0.04^\circ/h$, while the medium precision fiber-optic gyro is $\sim 0.4^\circ/h$. The dead zones are absolutely eliminated after modifying the modulator circuit boards and are comparable to the noise level, better than $0.001^\circ/h$ and $0.01^\circ/h$, respectively.

5 Conclusion

The mechanism of the dead zone in closed-loop IFOGs is theoretically analyzed. The digital closed-loop transfer model for fiber-optic gyroscope is modified and the electronic cross-coupling interference from the feedback channel to the output signal of the detector in the forward channel is added. Using this model, we achieve the real-time dynamic simulation for the dead zone. The results show that the dead zone is mainly caused by the electronic cross-coupling interference. According to the results, the circuit board has been redesigned to reduce the cross-coupling of electronic circuits. Before the electronic cross-coupling has been eliminated, the dead zone of the high-

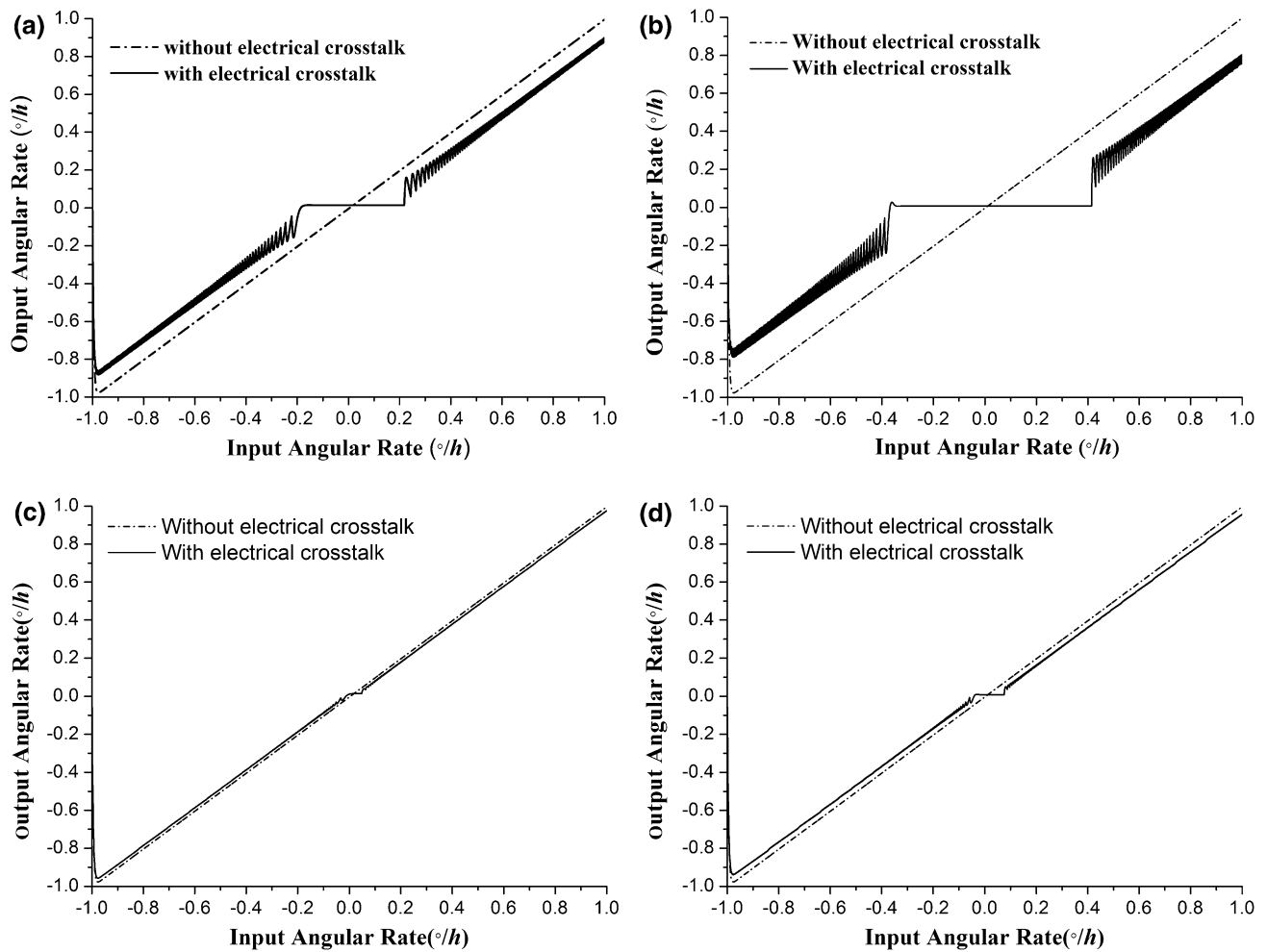


Fig. 6 Real-time dynamic simulation results and dead zone suppression of IFOGs in different conditions of fiber length, coil diameter, and electrical crosstalk. **a** $L = 1000$ m, $D = 100$ mm, $K_c = 5 \times 10^{-7}$; **b** $L = 1000$ m, $D = 100$ mm, $K_c = 1 \times 10^{-6}$;

c $L = 3000$ m, $D = 200$ mm, $K_c = 5 \times 10^{-7}$; **d** $L = 3000$ m, $D = 200$ mm, $K_c = 1 \times 10^{-6}$

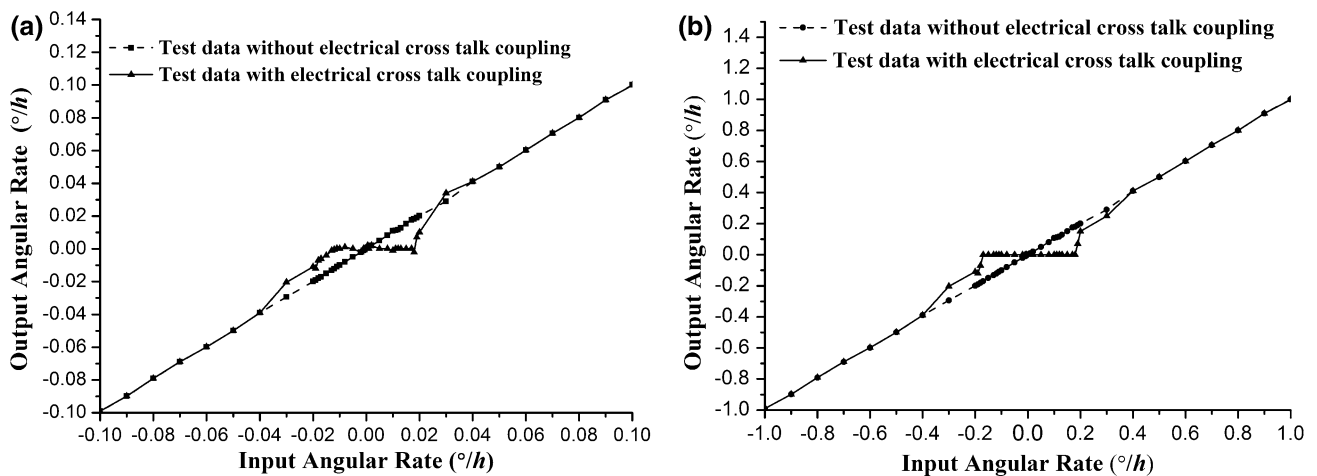


Fig. 7 Test data of the dead zone before and after electronic cross-coupling suppression. **a** $L = 3000$ m, $D = 200$ mm; **b** $L = 1000$ m, $D = 100$ mm

precision fiber-optic gyro is $\sim 0.04^\circ/h$, while the medium precision fiber-optic gyro is about $\sim 0.4^\circ/h$. After modifying the modulator circuit boards, the dead zones are absolutely eliminated and comparable to the noise level, better than $0.001^\circ/h$ and $0.01^\circ/h$, respectively.

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